

Start:

2:45

- Send agenda (working document saying what we want to talk about, issues, analysis, next step?)
<- graded based on comprehensiveness
- Each meeting we have a note taker
- Documents will end up on github in the end, but we can email them for now
- Can email whenever (2am) (best for emergencies) (piazza also works)

Potential questions (novel)

1. Does age have an impact on the relationship between depressive symptoms and gut microbiome of HIV patients?
2. Does immune status (HIV immune status) affect the association between depressive symptoms and the gut microbiome?
 - a. CD4/CD8 ratio is OK (maybe? **Look at literature to make sure**)
 - b. What about sex trends? Male vs female and should they be treated differently based on status?
 - c. Cant do glucose + ? (i forgot)
 - d. Greater chance that it had been done
3. What is the relationship between observed histopathology with metabolic markers and H. pylori presence?
 - a. H.Pylori: state limitation that H.pylori may not be present in sequencing data (just based on breath test)
 - b. Other bacteria's presence → histopathology
 - c. Gastric stage and age already done
 - d. I think we just scrap this one xd
4. Does H. pylori relate to loss of core microbiome stability (in terms of microbiome diversity) across the progression of gastric cancer?
 - a. Also doable
 - b. Is the microbiome altered significantly over progression(?)
 - c. Maybe use bar charts to compare the different groups
 - d. Would only be using people with diseases → leads to dataset being smaller
 - i. Small groups may lead to subsetting data to much
 - ii. Look at the number of samples in each group (metadata)
5. Does sex/gender modify the relationship between histological stage (HC→CG→IM→IN→GC) and gastric microbiome alpha diversity?
 - a. Sounds great (quite promising)
 - b. Sex difference under studied so good to focus on it
 - c. Reference to 2025 paper → they looked at age → follow up on that and look at sex

- i. Maybe first check if their results showed an actual difference. If they found nothing different, maybe better to do something else
 - ii. Read paper about histological stages from Ujemi
 1. Look at feasibility of the project from the paper (is looking at the stages worth looking at)
6. Are metabolic-risk markers associated with microbiome variation within the same histological stage?
 - a.
 - b.

His advice:

- Look at paper, check the stages to see if they are big enough for us to grow. If not interesting, maybe just do the metabolic markers because that hasn't been done before.
- Gastric cancer data may be better. Depression may have been done already.
- DO GOOD GOOGLE SCHOLAR SEARCH (Start broad and narrow down)

Literature search:

UJEMI article summary (<https://ojs.library.ubc.ca/index.php/UJEMI/article/view/200792>)

- Total sample size: 310
- Age range: 17-80
- Findings:
 1. Age **DOES NOT** significantly influence gastric microbiome(**alpha** AND **beta** diversity) dynamics during carcinogenesis
 2. **Older** patients microbiome does **not differ** across **different stages** of cancer
 3. **Younger** patients **lose** taxonomic profile with disease **progression** (looks more similar to older age groups) + **metabolic** ability is **suppressed** (evident by downregulation of biosynthetic & catabolic pathways + upregulation of L-1,2-propanediol degradation pathway)
 - Proteobacteria (healthy) -> Firmicutes (gastric cancer)
 - The upregulated pathway is in charge of producing key SCFAs. They hypothesized that although this is upregulated, in the context of everything else being suppressed, this is probably just a shift to a different carbon-using metabolic pathway (basically no overall increase in SCFAs).
- Methodology:
 - Made age into categorical data -> 45 young adult, 195 middle age(40-64), 70 elderly(65-80)
 - alpha and beta diversity metrics, core microbiome, indicator species, and predictive functional pathways were analyzed.
- Limitations

- The filtered + rarified sample size was fucked based on age (16 young adults, 121 middle aged, 41 elderly). This results in bias and skewing towards older populations.
- The young adult data lacked people with intestinal neoplasia + intraepithelial metaplasia -> couldn't fully compare between all stages of cancer
- Not generalizable:
 - Data from one geographical location + ethnicity
 - Diet was unavailable
 - Distribution of sexes for participants was not reported -> limits ability to do sex-specific effects → we do have sex in our data so we can use our study to fill in this potential knowledge gap
- Extra information:
 - Original paper:
 - H. pylori abundance does not differ across age or gastric cancer stage
 - Difference in gastric microbial diversity + composition doesn't rely on H. pylori
 - Functional composition differ as disease progresses

Google scholar search:

1. <https://link.springer.com/article/10.1186/s13293-025-00700-z>
 - a. Paper looked at sex specific effects of H. pylori (induced gastric cancer) on transgenic mice
 - b. They found that composition differs greatly between male and female REGARDLESS of H. pylori infection
 - c. **“males displayed an enrichment of pathogens, such as Escherichia Shigella and Streptococcus, females exhibited an over-representation of the beneficial Bifidobacterium”**
 - d. They used **16s rRNA** for analysis
 - e. Measured both alpha and beta diversity
 - Verdict: they used H. pylori induced gastric cancer. So if we just don't do H. pylori, then we should be fine. They also didn't look at specific stages of gastric cancer so we are fine (i think).
 - Also our dataset does not specify that the gastric cancer are H.pylori induced → we can still test for a difference between H.Pylori positive/negative and see if there is an effect because the paper doesn't test for H.pylori - patients
 - They also used mouse model and we have human samples

Metadata:

Sample sizes

Female: 126

- H.Pylori status: 86 negative + 40 positive
- Histopathology:
 - a. 22 Chronic gastritis (CG)
 - b. 32 Gastric cancer (GC)
 - c. 29 Healthy control (HC)
 - d. 22 Intestinal metaplasia
 - e. 21 Intraepithelial neoplasia (IN)

Male: 184

- H.Pylori status: 101 negative + 83 positive
- Histopathology:
 - a. 16 Chronic gastritis (CG)
 - b. 52 Gastric cancer (GC)
 - c. 30 Healthy control (HC)
 - d. 32 Intestinal metaplasia
 - e. 54 Intraepithelial neoplasia (IN)